Unit 5, Lesson 9: Finding Distances on a Coordinate Plane

Benchmark

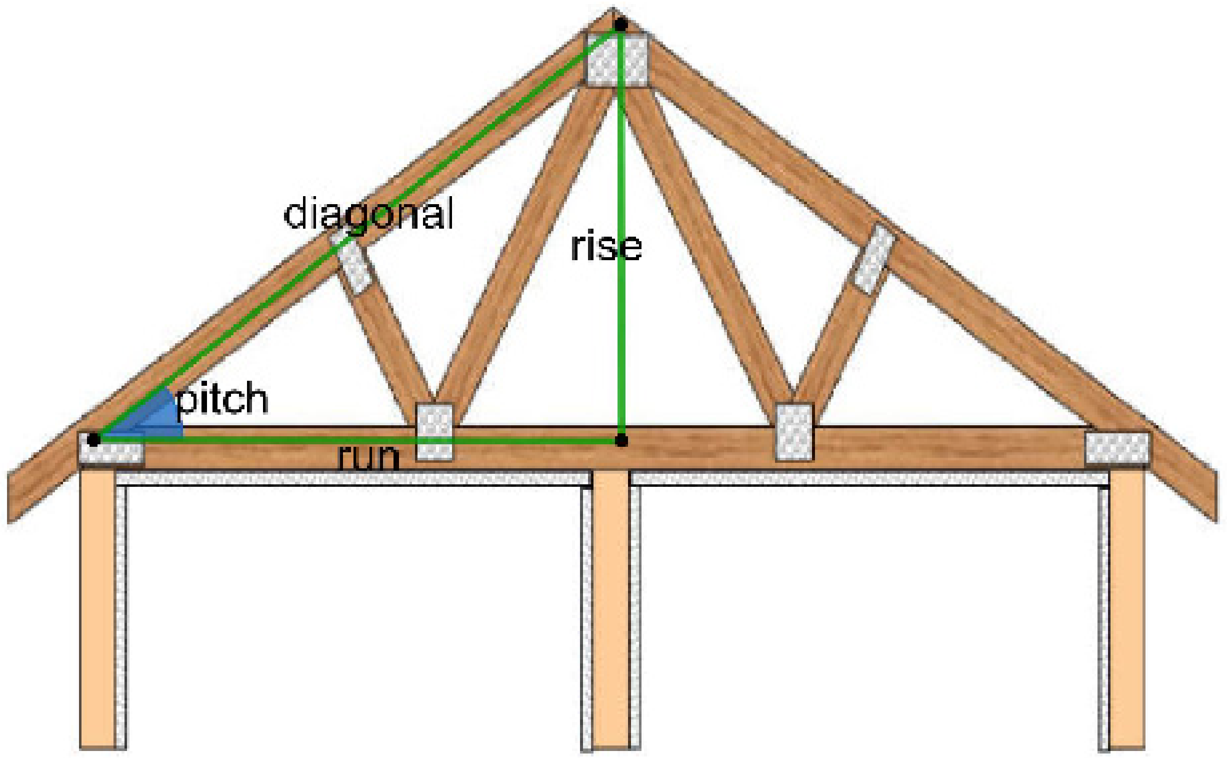
MA.8.GR.1.2 Apply the Pythagorean Theorem to solve mathematical and real-world problems involving the distance between two points in a coordinate plane.

Learning Target

- [ ] **I can determine the distance between two points on the coordinate plane.**

5.9.1 Warm-Up: Notice and Wonder

In carpentry, right triangles are commonly used in constructing roofs. A roof's pitch is the steepness of the roof. It can be determined by the ratio between the two legs of the right triangle.

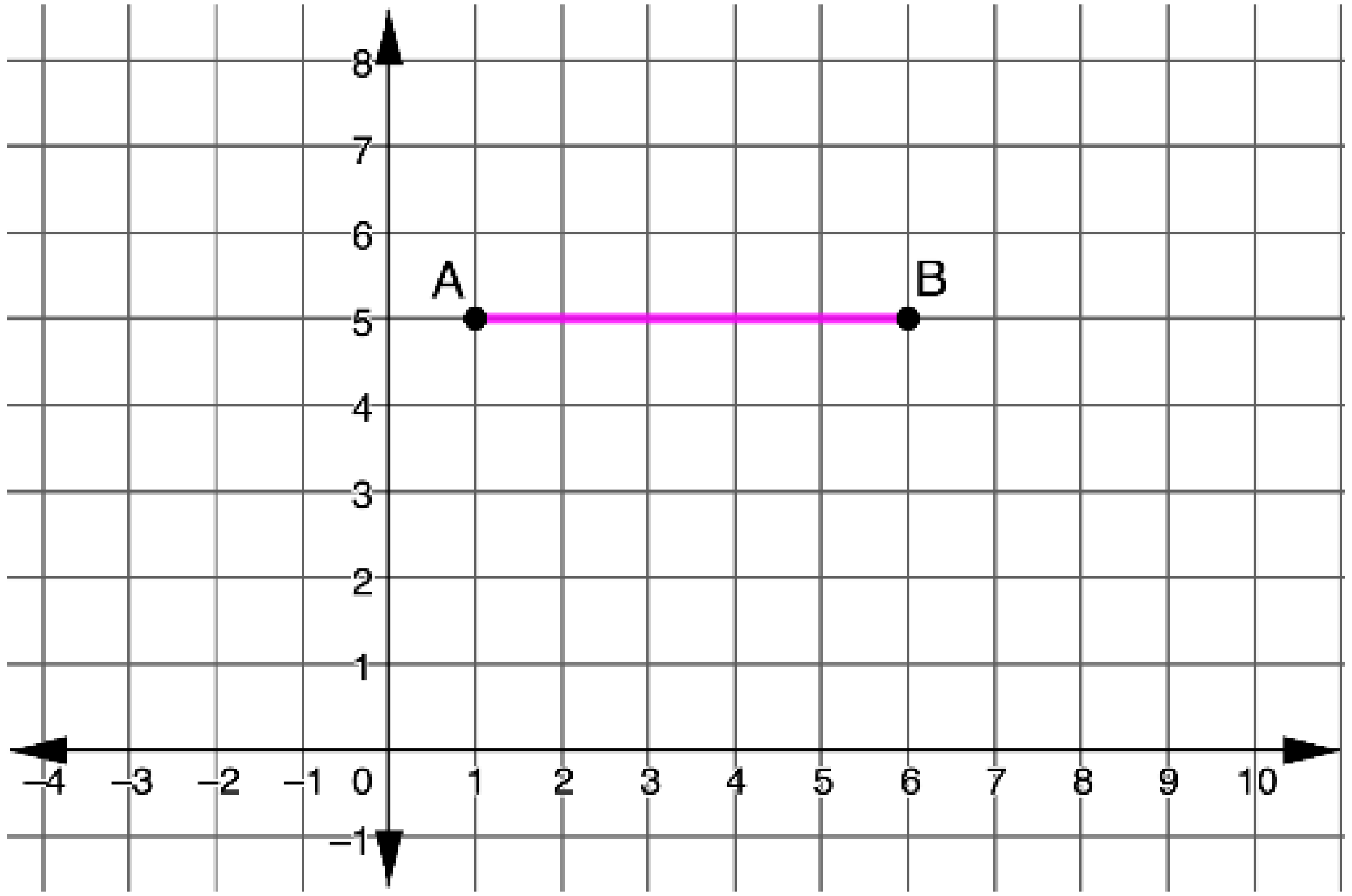


1. What do you notice or wonder about the use of right triangles in construction?

5.9.2 Exploration: Finding the Right Distance

Work with your partner to answer each of the following.

1. What is the distance between points A and B?



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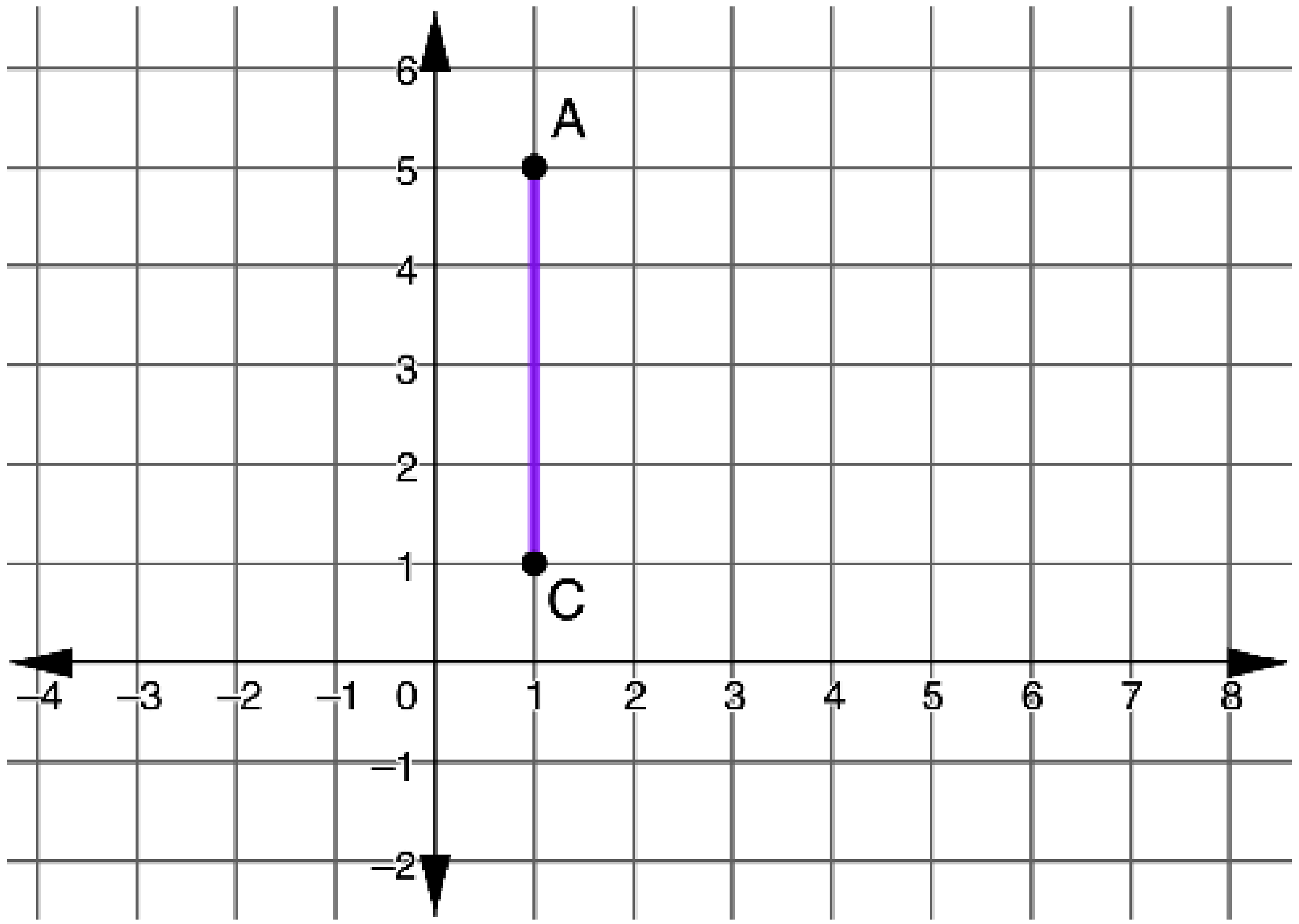
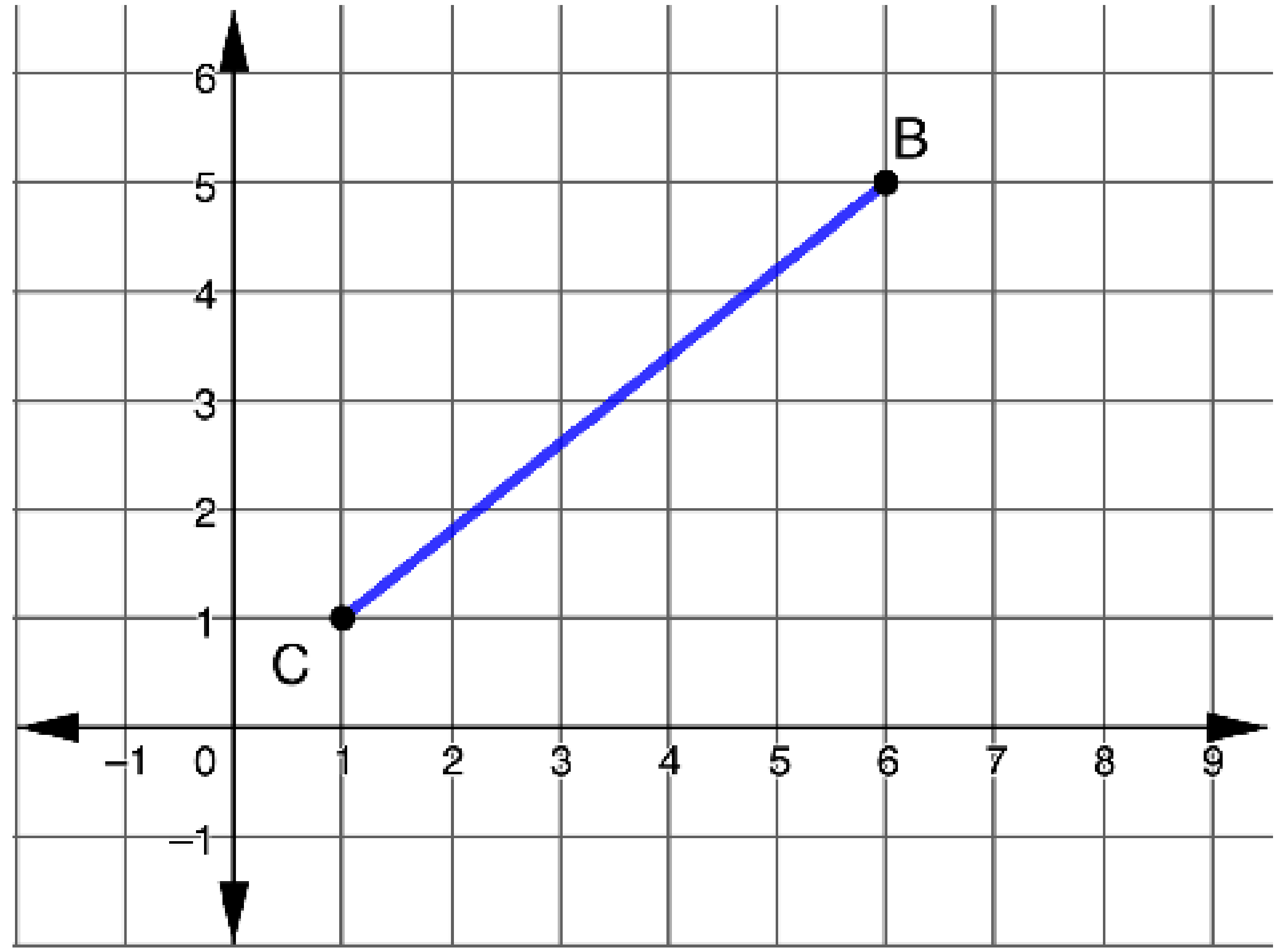
2. What is the distance between points A and C?



3. What strategy did you and your partner use to find the distance of the horizontal and vertical lines?

4. What is the distance between points B and C?



a. On the graph, label the length of segment BC, c.

b. Create a right triangle with hypotenuse c by sketching horizontal and vertical legs. Label the legs a and b.

c. Apply the Pythagorean Theorem to find the length of c. Round to the nearest tenth if necessary.

5. Discuss with your partner how to find the length of any diagonal on the coordinate plane.

6. Complete the statement.

The length of any non-vertical or non-horizontal segment can be determined by using the endpoints as a

- o hypotenuse
- o leg

of a right triangle and sketching the horizontal and

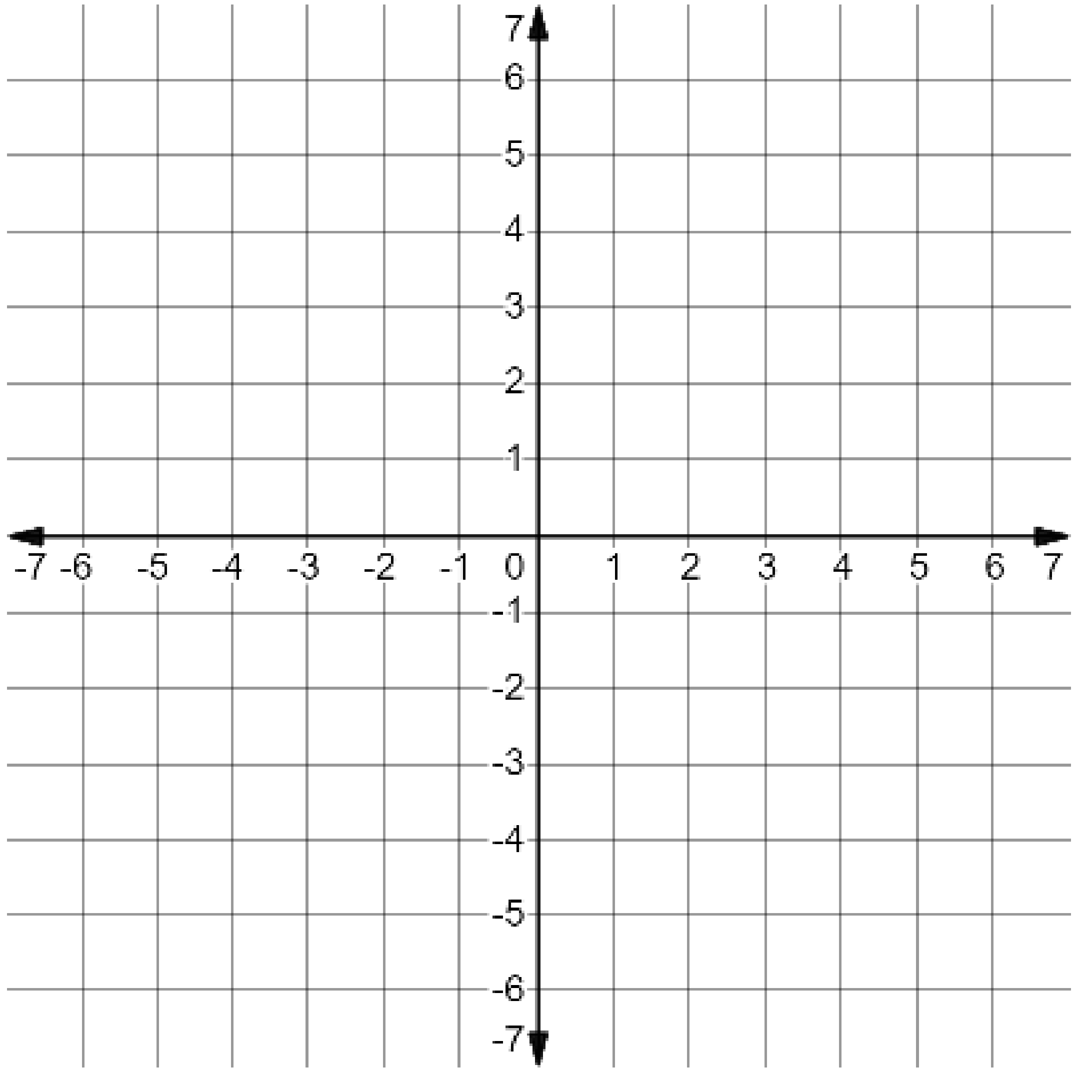
- o hypotenuse
- o legs

vertical then applying the

- o Pythagorean Theorem.
- o Area Formula.

5.9.3 Guided Practice: Finding Distance on the Coordinate Plane

1. Use the coordinate grid to find the exact distance between $(-3,1)$ and $(5,7)$.



a. Plot and connect the points on the grid.

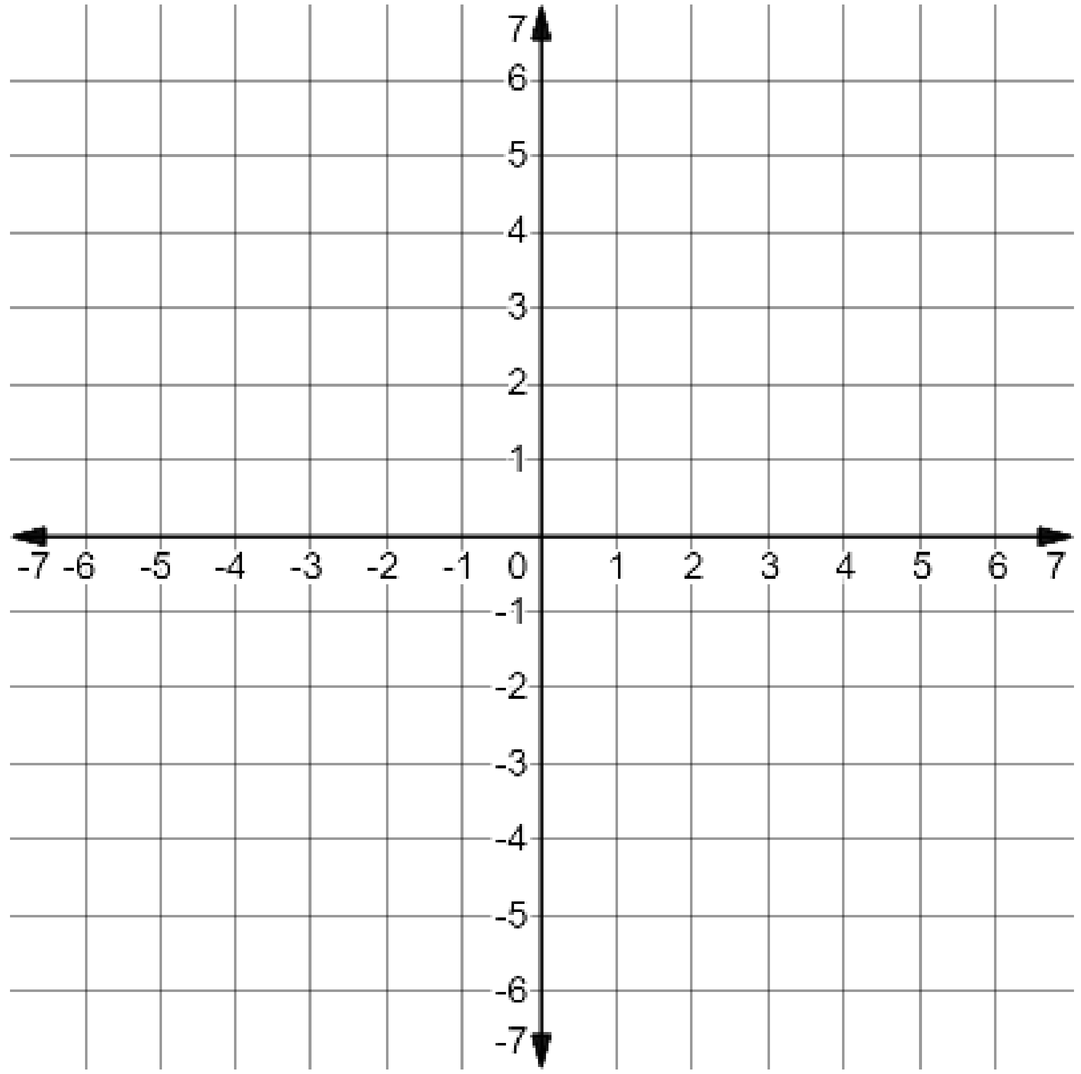
b. Create a right triangle by drawing horizontal and vertical line segments.

c. Apply the Pythagorean Theorem to find the missing side.

$$a^2 + b^2 = c^2$$

The exact distance between $(-3,1)$ and $(5,7)$ is __ units.

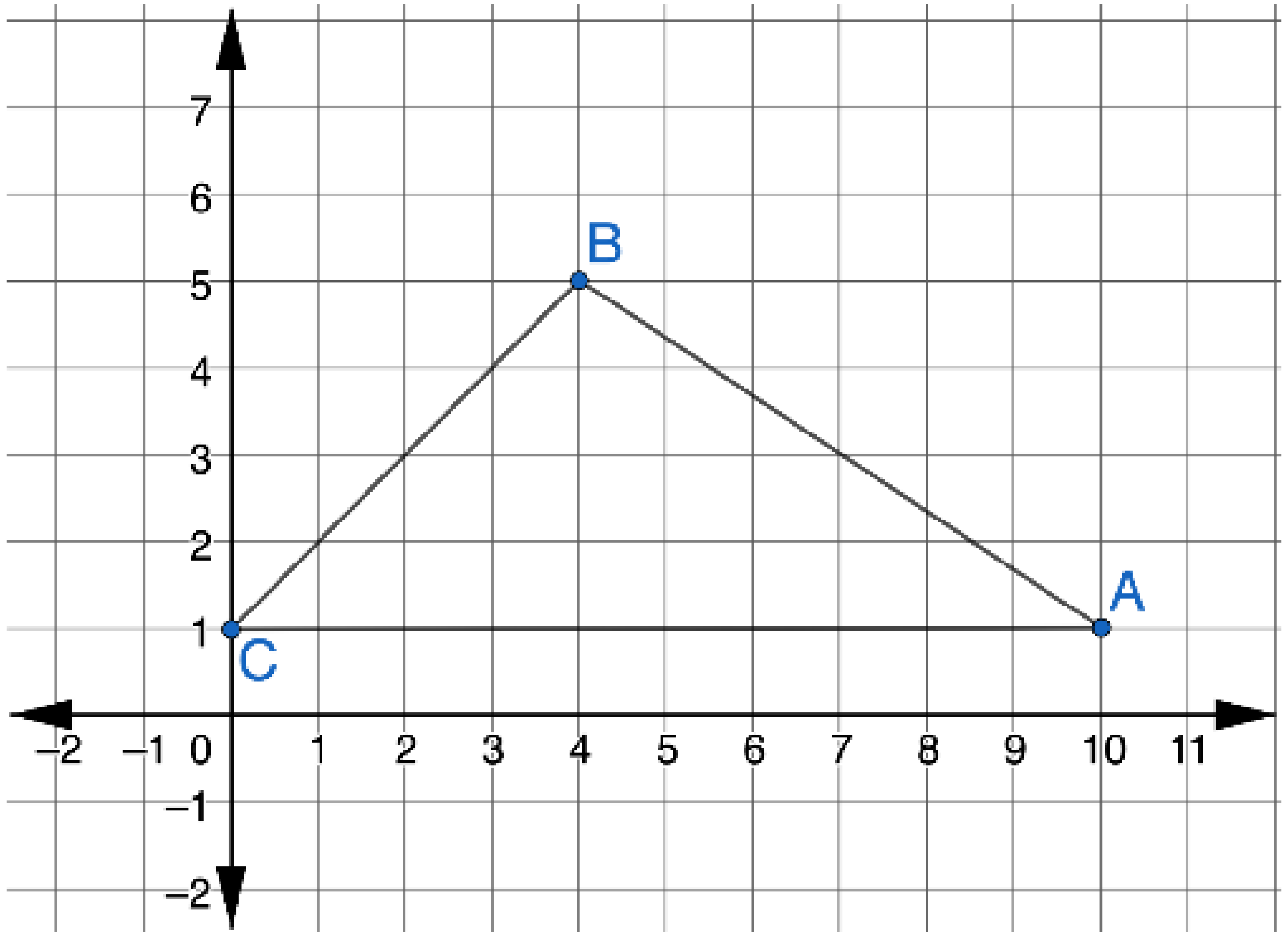
2. Use the coordinate grid to find the exact distance between $(-2,-5)$ and $(6,1)$.



The exact distance between $(-2,-5)$ and $(6,1)$ is __ units.

5.9.4 Guided Instruction: Converse of the Pythagorean Theorem

The sketch shows a design for a new roof.



1. Predict whether this design is in the shape of a right triangle.

2. To determine whether this is a right triangle, determine the length of each side and prove whether the Pythagorean Theorem holds true.

   a. Find the length of $\overline{AC}$.

b. Find the exact length of $\overline{AB}$ by creating a right triangle with $\overline{AB}$ as the hypotenuse.

c. Find the exact length of $\overline{BC}$ by creating a right triangle with $\overline{BC}$ as the hypotenuse.

d. List the three side lengths of the triangle.

$\overline{AC}$ = __________ units

$\overline{AB}$ = __________ units

$\overline{BC}$ = __________ units

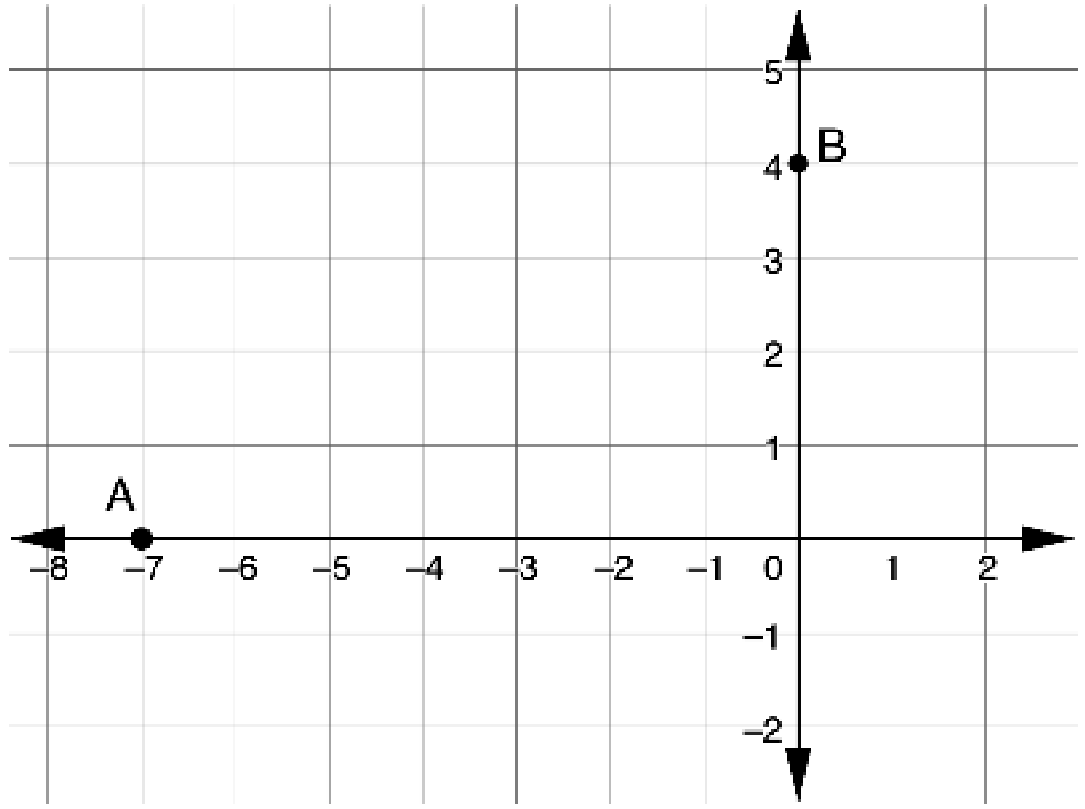
e. Determine if Triangle ABC is a right triangle using the converse of the Pythagorean Theorem.

3. Complete the statement.

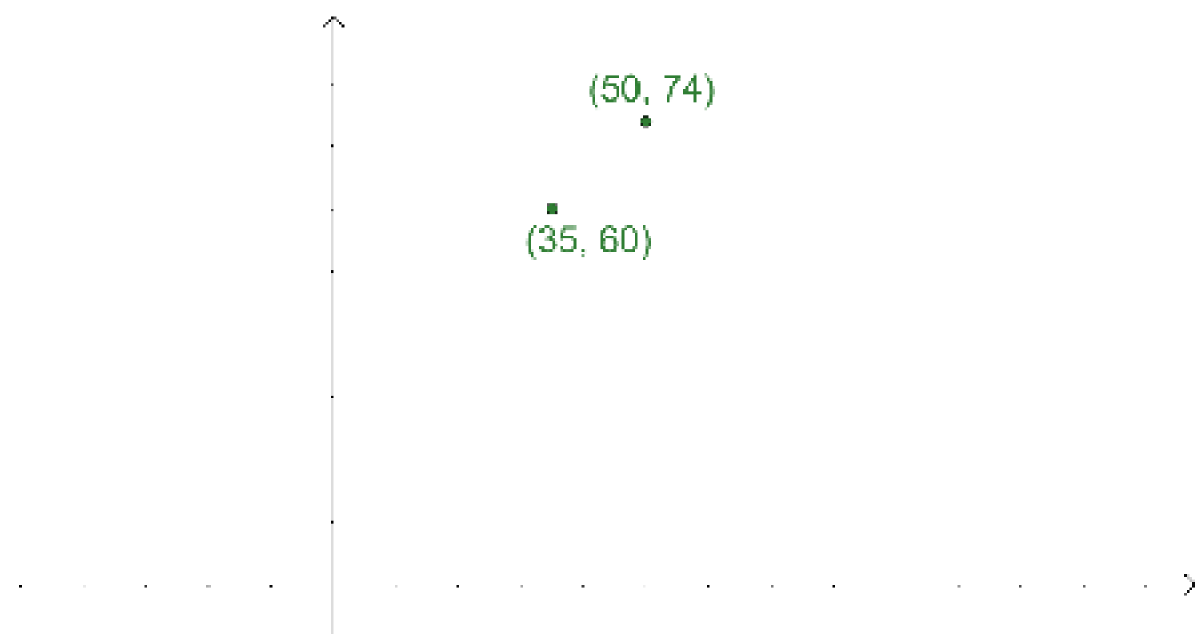
The roof design satisfy [○ does ○ does not] the Pythagorean Theorem, therefore, the [○ is ○ is not] design a right triangle.

5.9.5 Your Turn!

1. Find the exact distance from Point A to Point B.



2. Find the exact distance between the two points.



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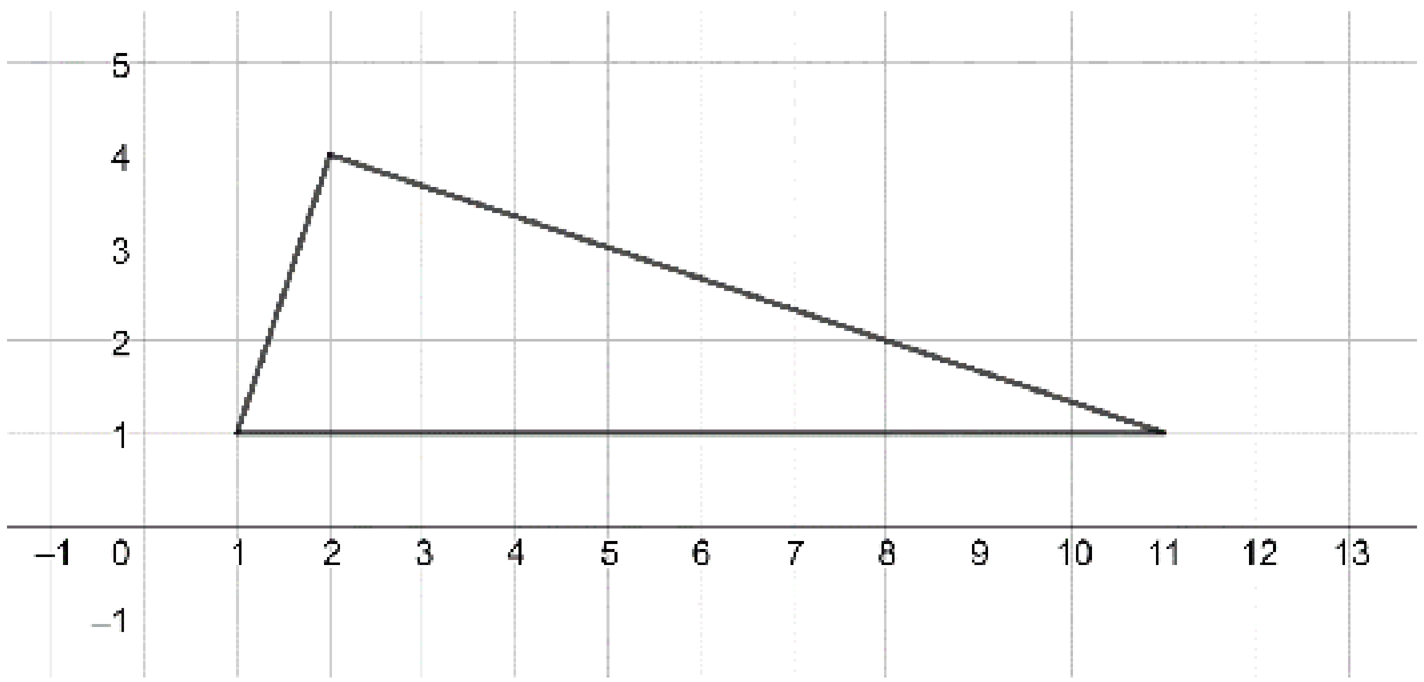
3. **Use the converse of the Pythagorean Theorem to determine if the figure is a right triangle.**



5.9.6 Wrap-Up: Reflection

1. **Select the emoji face that corresponds to your feeling toward today's learning target.**

I can determine the distance between two points on the coordinate plane.